

Notes: Schmitz (JPE, 2005)

What Determines Productivity? Lessons from the Dramatic Recovery of the U.S. and Canadian Iron Ore Industries Following Their Early 1980s Crisis

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1 Big picture

- **Question.** What determines productivity, and can a sudden increase in product-market competition force large productivity gains inside existing plants rather than only through selection among firms?
- **Empirical episode.** For roughly a century, the Great Lakes iron ore producers were insulated from foreign competition because transportation costs made local ore the cheapest supply for Great Lakes steel producers. In the early 1980s, changes in the world steel and shipping environment allowed Brazilian ore to be delivered to Chicago at prices below local ore.
- **Crisis.** The U.S. and Canadian iron ore industries suddenly faced a credible threat to their core market. The threat was not a slow trend; it was a large competitive shock that put the survival of the regional industry into question.
- **Main result.** Labor productivity in the U.S. and Canadian iron ore industries roughly doubled within a few years. In Canada, materials productivity increased by more than half, capital productivity rose, and total factor productivity increased sharply.
- **Mechanism.** Most of the productivity gains came from changes in work practices. Restrictive work rules had produced overstaffing and kept machines in nonproduction or repair mode for significant periods. Loosening those rules increased output per worker, output per unit of capital, and output per unit of materials.
- **Selection versus within-plant improvement.** The paper shows that the productivity boom was not primarily a mechanical reallocation toward high-productivity mines. A formal decomposition for Minnesota pellet mines shows that within-mine productivity growth explains the dominant share of the gain.
- **Technology versus work rules.** The evidence does not support a simple explanation based on capital deepening, substitution away from expensive labor, or a new equipment technology. The strongest evidence points to labor-rule changes that made existing capital operate more productively.
- **Incremental contribution.** The paper provides a rare, industry-level natural experiment in which the source of productivity growth can be traced from macro competition to plant-level work practices, rather than stopping at reduced-form correlations between competition and productivity.
- **Economic takeaway.** Competition matters not only because it reallocates market share to efficient producers. It can also discipline organizations internally by breaking inefficient work practices and changing how existing resources are used.

2 Incremental contribution of the paper

2.1 Contribution relative to the competition-productivity literature

Schmitz studies a setting where the competitive shock is unusually clear. The Great Lakes iron ore industry was protected by geography, not by a formal tariff, and the protection abruptly weakened when foreign suppliers became credible competitors in the Great Lakes steel market. This creates an industry-level experiment that is easier to interpret than gradual entry, deregulation, or cross-sectional comparisons of concentration.

- Existing work had often shown that productivity is higher where competition is stronger, but it was difficult to identify exactly what changed inside firms.
- This paper goes further by documenting the organizational channel: work practices changed, overstaffing declined, and equipment operated more hours.
- The paper therefore connects competition to a concrete production mechanism rather than treating productivity as a black-box residual.

2.2 Contribution relative to selection-based explanations

A common view is that competition increases industry productivity mostly by forcing inefficient firms to exit. Schmitz shows that this cannot be the whole story here.

- Some mines did close, and some market-share reallocation occurred.
- But the formal decomposition in Table 2 shows that continuing mines' own productivity improvements account for most industry labor-productivity growth.
- This demonstrates a within-unit productivity channel: competition can raise productivity by changing how incumbents operate, even before or apart from exit.

2.3 Contribution relative to technology/capital-deepening explanations

The paper also separates productivity gains from standard neoclassical substitution stories.

- Figures 8 and 9 compare relative factor prices with input ratios.
- The timing and magnitude of the productivity boom do not look like a standard substitution response to rising labor costs.
- Productivity gains occur when work practices change and when repair/nonproduction time declines, not simply when capital-labor or electricity-labor ratios adjust.

2.4 Contribution to organizational economics

The paper provides evidence that restrictive work practices can be a major determinant of measured productivity.

- The mechanism is organizational: who is allowed to do which tasks, how quickly machines can be restarted, how staffing is allocated, and how much time equipment spends producing output.
- The same physical capital and broad technology can generate very different productivity levels under different work rules.
- This connects macro productivity outcomes to micro contracting and labor-relations institutions.

2.5 Contribution to historical industrial organization

The paper documents an episode where geography functioned like a protective trade barrier.

- Large transport costs protected Great Lakes ore producers for decades.
- The early-1980s foreign competition shock was analogous to a sudden tariff cut.
- The paper therefore contributes to trade and industrial organization by showing how a decline in effective protection can transform producer behavior.

3 Data and measurement

3.1 Industry setting and production units

The paper focuses on iron ore producers supplying the Great Lakes steel market. The main units of observation are the U.S. and Canadian iron ore industries, and in some sections individual Minnesota pellet mines. The U.S. analysis emphasizes the Minnesota pellet industry because the pellet mines were central to Great Lakes ore production and because mine-level data are available.

- **Output.** Output is measured in tons of iron ore or pellets.
- **Labor.** Labor input is hours worked, allowing labor productivity to be measured as tons per hour.
- **Materials.** Materials include electricity and fuel, supplies, grinding media, repair parts, and contract work.
- **Capital.** Capital productivity is measured through output relative to capital services or capital stock proxies.
- **Mine-level records.** Minnesota pellet mine data allow decomposition of industry productivity into within-mine, between-mine, cross, closing, and entering components.

Interpretation: The key empirical advantage is that the paper can connect aggregate productivity patterns to actual mine-level production, hours, and work-practice changes.

3.2 Productivity concepts

The paper uses several productivity measures.

- **Labor productivity:** output per hour, Y_t/N_t .
- **Materials productivity:** output per unit of materials, Y_t/M_t .
- **Capital productivity:** output per unit of capital, Y_t/K_t .
- **Total factor productivity:** output relative to a weighted bundle of labor, materials, and capital.

A stylized TFP measure can be written as

$$TFP_t = \frac{Y_t}{N_t^{s_N} M_t^{s_M} K_t^{s_K}},$$

where the s terms are factor-cost shares.

Interpretation: Labor productivity alone can rise because of capital deepening or materials deepening. The Canadian TFP evidence is important because it shows that the productivity increase was broader than output per worker alone.

3.3 Work-practice evidence

The paper uses contract evidence, industry interviews, and mine-level records to identify changes in work practices. The central examples involve rules governing repair workers, machine operators, job classifications, overstaffing, and the ability to restart or repair equipment quickly.

- Restrictive work rules created staffing requirements not tightly linked to production needs.
- They also meant machines could remain idle while waiting for workers with the correct classification.
- When crisis bargaining loosened those rules, fewer workers could operate more output-producing time.

Interpretation: The mechanism matters because it explains why labor, materials, and capital productivity can rise together. If the same machines spend more time producing, output rises relative to all three major inputs.

3.4 Competition shock

The paper treats the early-1980s foreign competition threat as the key shock.

- Great Lakes producers had been protected by location and transport costs.
- Brazilian producers became able to supply Chicago-area steel producers at lower prices.
- This shock occurred because of broader world steel market developments, not because the Great Lakes producers voluntarily changed their technology.

Interpretation: The shock is analogous to a large reduction in effective trade protection. Its timing provides leverage for interpreting the productivity acceleration as a competitive response.

4 Models, accounting decompositions, and mechanisms

4.1 Industry labor-productivity decomposition

Let $\pi_{it} = y_{it}/n_{it}$ be labor productivity at mine i , $s_{it} = n_{it}/N_t$ be the mine's share of industry hours, and $\Pi_t = Y_t/N_t = \sum_i s_{it}\pi_{it}$ be industry labor productivity. The change from base year t to year t' can be decomposed as

$$\Delta\Pi_{t,t'} = \sum_{i \in C} s_{i,t} \Delta\pi_i + \sum_{i \in C} (\pi_{i,t} - \Pi_t) \Delta s_i + \sum_{i \in C} \Delta\pi_i \Delta s_i - \sum_{i \in X} s_{i,t} (\pi_{i,t} - \Pi_t) + \sum_{i \in E} s_{i,t'} (\pi_{i,t'} - \Pi_t).$$

- The first term is within-mine productivity growth.
- The second term captures reallocation of hours toward initially high-productivity mines.
- The third term is the cross term between productivity changes and share changes.
- The fourth term captures the effect of exits.
- The fifth term captures the effect of entrants.

Interpretation: If competition works mostly through selection, the between and exit terms should dominate. In Table 2, the within-mine term dominates, which points to changes in how continuing mines operated.

4.2 Work-practice production mechanism

The appendix formalizes the idea that a mine's output depends not only on inputs, but also on the fraction of time equipment is in operating mode. A simplified version is

$$Y = A \cdot \frac{a}{[a + (1-a)m]^{\nu_m}} N_p^{\nu_n} M^{\nu_m} K^{\nu_k},$$

where a is the fraction of time the mine is operating, m is materials consumed during nonoperation relative to operation, N_p is productive labor, M is materials, and K is capital.

Interpretation: Loosening work practices can increase a and reduce nonproductive labor. This raises measured labor productivity, materials productivity, and capital productivity even without a large change in physical technology.

4.3 Factor-substitution alternatives

A standard substitution story would predict that firms use less labor relative to capital or energy when labor becomes more expensive. The paper checks this by comparing relative input prices with relative input usage.

- If capital-labor substitution drove the productivity gain, capital per worker should move strongly with the relative price of labor to capital.
- If energy-labor substitution drove the productivity gain, electricity per worker should move strongly with the relative price of labor to electricity.
- Figures 8 and 9 do not support these stories as the main explanation.

Interpretation: The productivity change is not primarily a smooth cost-minimizing movement along a production function. It is an organizational change in how production was carried out.

5 Identification and empirical design

5.1 What identifies the competition effect

The paper's identification is historical and event-based. It uses a large, externally driven shock to the competitive environment faced by Great Lakes iron ore producers.

- The competitive threat arrived abruptly in the early 1980s.
- Productivity had been mostly flat before the shock.
- Productivity rose sharply after the shock.
- The threat came from foreign suppliers, especially Brazil, rather than from an internal mine-level technology program.

Interpretation: The design is not a randomized experiment, but the timing and magnitude of the competitive shock provide a strong historical contrast.

5.2 Why within-mine evidence is central

A concern is that industry productivity might rise simply because low-productivity mines close. The mine-level decomposition addresses this concern.

- Table 2 shows that continuing mines account for most productivity growth.
- Closing mines and reallocation contribute much less than within-mine improvement.
- Figure 7 confirms that multiple continuing mines improved productivity after the crisis.

Interpretation: This is the strongest evidence that competition changed behavior inside firms, not just the composition of firms.

5.3 Why work-practice evidence is credible

The paper triangulates the work-practice mechanism using several forms of evidence.

- Industry narratives and contract evidence describe changes in restrictive work rules.

- Figure 10 shows a direct decline in repair hours at Minntac/USX.
- Figures 5 and 6 show materials and capital productivity rising, consistent with equipment spending more time operating.
- Appendix theory explains why the same work-rule change can raise labor, capital, materials, and TFP measures simultaneously.

Interpretation: The mechanism is persuasive because the qualitative institutional evidence and the quantitative productivity patterns line up.

5.4 Threats and limitations

- The paper cannot perfectly separate all simultaneous changes in management, technology, labor relations, and demand.
- Some productivity gains could reflect learning, managerial effort, or mine-specific capital adjustments not fully observed.
- The materials and capital data are stronger for Canada than for the United States.
- The setting is a heavy industrial case study, so external validity to all industries requires caution.

Interpretation: The paper is strongest as a detailed industry case showing one concrete mechanism by which competition can raise productivity.

6 Section-by-section study guide

- **Introduction.** Establishes the core puzzle: protected Great Lakes producers faced sudden foreign competition and productivity surged.
- **Precrisis protection.** Explains why transport costs historically insulated the region from foreign ore.
- **Crisis.** Shows why the early-1980s world steel shock made foreign ore a credible competitor.
- **Productivity facts.** Documents labor, materials, capital, and TFP increases in U.S. and Canadian iron ore.
- **Decompositions.** Separates within-mine productivity gains from reallocation, closures, and entry.
- **Work practices.** Provides the central explanation for the productivity surge.
- **Factor-substitution checks.** Shows that conventional input-substitution stories do not explain the timing or magnitude of the gains.
- **Appendices.** Provide materials measurement and a formal model of how work practices affect productivity.

7 Tables and figures

7.1 Figure 1: Pig iron production by various regions, 1950–96 + Figure 2: Production and labor productivity: Minnesota pellet industry

Read:

- Figure 1 shows the long-run production environment around the Great Lakes iron ore market.
- U.S. pig iron production declines sharply after the late 1970s and early 1980s, signaling a major shock to local ore demand.
- Canada and other regions are much smaller in absolute production, while Europe and the United States undergo large changes.
- Figure 2 shows the Minnesota pellet industry before and after the crisis.
- Production falls after the early-1980s crisis, but labor productivity rises sharply from the mid-1980s onward.
- Labor productivity roughly doubles while production remains below earlier peaks.

Detailed interpretation: Figure 1 provides the market background: the Great Lakes ore industry is hit by a shock associated with world steel-market upheaval. Figure 2 provides the key industry-level fact: the local industry did not merely shrink; it reorganized and produced much more output per hour.

Economic message: The first pair of figures establishes the paper’s empirical puzzle: a crisis caused by foreign competition was followed by a striking productivity recovery.

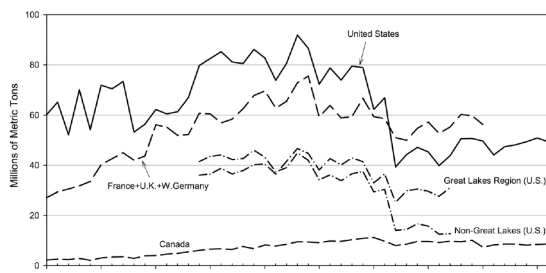


FIG. 1.—Pig iron production by various regions, 1950–96

Figure 1: Pig iron production by region



FIG. 2.—Production and labor productivity: Minnesota pellet industry

Figure 2: Minnesota pellet industry production and labor productivity

7.2 Figure 3: Production and labor productivity: U.S. iron ore industry + Figure 4: Production and labor productivity: Canadian iron ore industry

Read:

- Figure 3 shows that U.S. iron ore labor productivity rises sharply after the early-1980s downturn.
- Output falls around the crisis, while output per hour rises strongly afterward.
- Figure 4 shows a very similar pattern for Canada.
- The Canadian industry also experiences a large labor-productivity increase after the competitive shock.
- The similarity across the two countries suggests that the productivity response was not idiosyncratic to a single mine or firm.

Detailed interpretation: The U.S. and Canadian patterns reinforce each other. Two neighboring industries, both historically linked to Great Lakes steel and both exposed to the foreign ore threat, show the same postcrisis productivity takeoff.

Economic message: Competition produced a broad regional productivity response, not an isolated mine-level anomaly.



FIG. 3.—Production and labor productivity: U.S. iron ore industry

Figure 3: U.S. iron ore industry

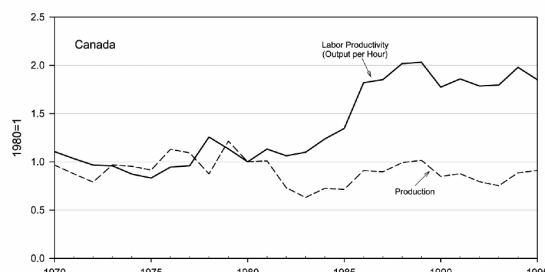


FIG. 4.—Production and labor productivity: Canadian iron ore industry

Figure 4: Canadian iron ore industry

7.3 Figure 5: Materials productivity and capital productivity: Canadian iron ore industry + Figure 6: Total factor productivity: Canadian iron ore industry

Read:

- Figure 5 shows that Canadian materials productivity rises substantially after the crisis.
- Capital productivity also rises, although less dramatically than materials productivity.
- Figure 6 decomposes Canadian TFP into labor productivity and the quantities used in the calculation.
- TFP rises sharply, though not as much as labor productivity.
- Materials and capital terms do not move in a way that would explain the whole labor-productivity jump by capital or materials deepening.

Detailed interpretation: These figures are crucial because they show the productivity boom is not only a labor-hours story. Output increases relative to materials and capital too, consistent with more effective operation of the same production system.

Economic message: The evidence points toward organizational efficiency: machines and materials were used more productively when work practices changed.

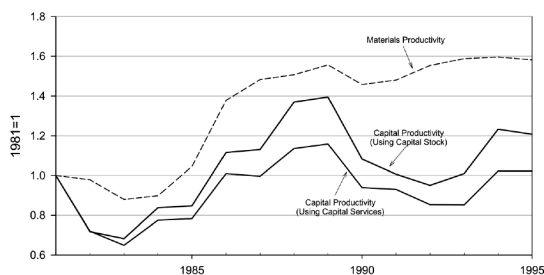


FIG. 5.—Materials productivity and capital productivity: Canadian iron ore industry

Figure 5: Canadian materials and capital productivity

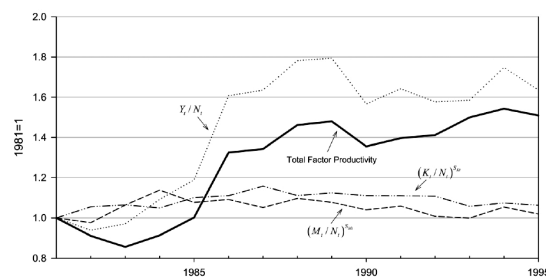


FIG. 6.—Total factor productivity (and quantities used in its calculation): Canadian iron ore industry

Figure 6: Canadian TFP and components

7.4 Table 1: Total Factor Productivity: Canadian Iron Ore Industry + Figure 7: Production and labor productivity: Minnesota pellet mines

Read:

- Table 1 reports Canadian TFP rising from 1.00 in 1981 to 1.51 in 1995.
- Output per hour rises even more, from 1.00 in 1981 to 1.64 in 1995.
- Materials and capital components are comparatively stable, which means the TFP increase reflects more than simple input deepening.
- Figure 7 shows that many individual Minnesota pellet mines experienced large productivity increases.
- Some mines close or temporarily stop operating, but the rise is visible among continuing operations as well.

Detailed interpretation: Table 1 quantifies the aggregate Canadian productivity increase, while Figure 7 shifts attention to the mine level. Together they show that the productivity boom is both aggregate and micro-level.

Economic message: The paper's mechanism requires continuing production units to improve, and these visuals support that requirement.

YEAR	TOTAL FACTOR PRODUCTIVITY	CALCULATED FROM		
		Y/N_t	$(M_t/N_t)^{\%}$	$(K_t/N_t)^{\%}$
1981	1.00	1.00	1.00	1.00
1982	.91	.94	.98	1.06
1983	.86	.97	1.06	1.07
1984	.91	1.09	1.14	1.05
1985	1.00	1.19	1.08	1.10
1986	1.33	1.61	1.09	1.11
1987	1.34	1.64	1.05	1.16
1988	1.46	1.78	1.10	1.11
1989	1.48	1.79	1.08	1.12
1990	1.36	1.57	1.04	1.11
1991	1.40	1.64	1.06	1.11
1992	1.41	1.58	1.01	1.11
1993	1.50	1.59	1.00	1.06
1994	1.54	1.75	1.05	1.07
1995	1.51	1.64	1.02	1.06

Table 1: Canadian TFP



Figure 7: Minnesota pellet mines

7.5 Table 2: Decomposition of Industry Labor Productivity Growth: Minnesota Pellet Industry

Read:

- Table 2 decomposes productivity growth relative to 1980 into within-mine, between-mine, cross-mine, closing-mine, and entrant components.
- The within-mine component is consistently the largest source of productivity gains.
- By the late 1980s and 1990s, within-mine gains explain most of the total industry labor-productivity increase.
- Closing mines contribute some gains in selected years, but they are not the dominant source.
- Entry contributes essentially nothing because there are no new mines in the period.

Detailed interpretation: This table is one of the paper's most important results. It rejects the interpretation that productivity rose mainly because low-productivity mines exited. Instead, productivity rose because existing mines changed how they operated.

Economic message: Competition can raise productivity through internal reorganization, not merely by reallocating market share across producers.

TABLE 2
DECOMPOSITION OF INDUSTRY LABOR PRODUCTIVITY GROWTH: MINNESOTA PELLET
INDUSTRY

GROWTH BETWEEN 1980 AND	OVERALL INDUSTRY GROWTH (1)	SHARE OF INDUSTRY GROWTH DUE TO			
		Within Mines (2)	Between Mines (3)	Cross Mines (4)	Closing Mines (5)
1981	10.20	105	-16	11	0
1982	0	...	...	...	...
1983	13.60	79	16	5	0
1984	55.10	93	6	1	0
1985	67.90	97	3	0	0
1986	77.50	87	7	6	0
1987	121.50	77	3	14	6
1988	108.80	76	3	15	7
1989	101.80	73	3	16	7
1990	100.90	95	7	-2	0
1991	87.20	96	9	-5	0
1992	91.70	92	9	-1	0
1993	104.40	108	6	-13	0
1994	113.70	106	6	-12	0
1995	119.90	101	6	-7	0

NOTE.—All figures are percentages. Ellipses in place of a figure mean that it is not defined (growth was zero between 1980 and 1982). Weights are the mine's share of industry hours.

Table 2: Decomposition of Minnesota pellet industry labor productivity growth

7.6 Figure 8: Price of labor relative to price of new capital vs. capital usage relative to labor usage + Figure 9: Price of labor relative to price of electricity vs. electricity usage relative to labor usage

Read:

- Figure 8 compares the labor-to-capital relative price with capital per unit of labor in Canada.
- Figure 9 compares the labor-to-electricity relative price with electricity per unit of labor.
- If productivity growth were primarily driven by input substitution, usage ratios should move closely with relative prices.
- The patterns do not line up well enough to explain the productivity surge.
- The figures therefore weaken a standard capital-deepening or energy-substitution explanation.

Detailed interpretation: These figures are diagnostic. They do not deny that input ratios move, but they show that the productivity boom is not well explained by conventional relative-price substitution.

Economic message: The evidence narrows the mechanism: the main force is not simply using more capital or electricity per worker; it is changing work practices and operating efficiency.

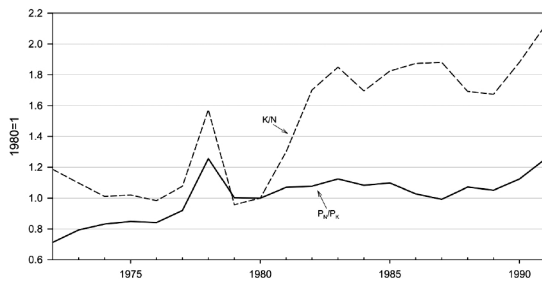


FIG. 8.—Price of labor relative to price of new capital vs. capital usage relative to labor usage: Canadian iron ore industry

Figure 8: Labor price vs. capital/labor use



FIG. 9.—Price of labor relative to price of electricity vs. electricity usage relative to labor usage: Canadian iron ore industry

Figure 9: Labor price vs. electricity/labor use

7.7 Figure 10: Total hours and repair hours as a percentage of total hours: Minntac/USX pellet mine + Figure 11: Production and labor productivity: Robe River iron ore

Read:

- Figure 10 shows that repair hours as a share of total hours decline sharply at Minntac/USX after the crisis period.
- Total hours also drop substantially, especially around the mid-1980s.
- The decline in repair-hour intensity is direct evidence that staffing and repair practices changed.
- Figure 11 shows Robe River in Australia, an industry not facing the same competitive shock, with a very different timing of productivity growth.
- Robe River's productivity rises after its own work-practice and operational changes, reinforcing the mechanism that work practices can drive productivity.

Detailed interpretation: Figure 10 provides the micro evidence for the paper's work-practice mechanism. Figure 11 provides an external comparison: large productivity changes arise when production organization changes, not necessarily when product demand changes alone.

Economic message: The production process itself matters. Work rules can determine how much of the day capital actually produces output.

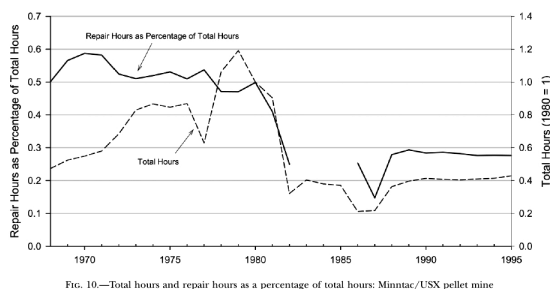


Figure 10: Minntac/USX repair hours and total hours

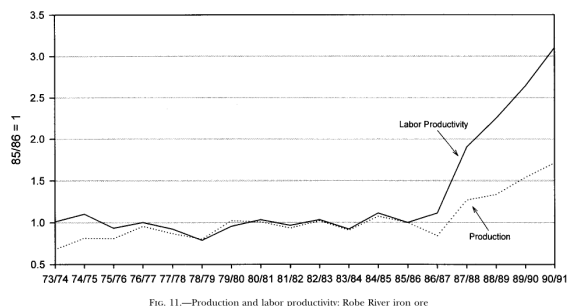


Figure 11: Robe River production and labor productivity

7.8 Table B1: Materials Spending in the U.S. Iron Ore Industry, 1982 and 1987

Read:

- Table B1 breaks U.S. iron ore materials spending into electricity and fuel, supplies, and contract work.
- Electricity and fuel are the largest components, accounting for more than half of materials spending in both 1982 and 1987.
- The materials composition is fairly stable, which helps the author construct a materials series despite data limitations.
- The table supports the appendix calculation that U.S. materials productivity increased, though less dramatically than in Canada.

Detailed interpretation: The appendix table matters because materials productivity is harder to measure than labor productivity. The table shows that the industry had a detailed and relatively stable materials cost structure, giving credibility to the materials-productivity calculation.

Economic message: The materials evidence supports the broader point that work-practice changes improved efficiency across multiple inputs, not just labor.

TABLE B1
MATERIALS SPENDING IN THE U.S. IRON ORE INDUSTRY, 1982 AND 1987

	1982		1987	
	Spending (\$)*	Total Materials (%)	Spending (\$)*	Total Materials (%)
Electricity and fuel:				
Electricity	252.2	38.3	258.1	42.8
Fuel for vehicles	29.3	4.5	14.3	2.4
Fuel for furnaces	85.8	13.0	51.3	8.5
Total	367.3	55.8	323.7	53.7
Supplies:				
Explosives	14.9	2.3	13.6	2.3
Repair parts	42.0	6.4	48.0	8.0
Bentonite	17.1	2.6	16.1	2.7
Metallic grinding media	36.2	5.5	50.2	8.3
Other supplies	161.0	24.5	117.1	19.4
Total	271.2	41.3	245.0	40.7
Contract work	19.2	2.9	34.0	5.6
Total materials	657.7	100.0	602.7	100.0

* Dollar figures are given in millions of current dollars.

Table B1: U.S. iron ore materials spending, 1982 and 1987

8 Detailed result map

8.1 Result 1: The crisis was a competition shock

- For decades, Great Lakes producers were protected by transport costs.
- The early-1980s shock made foreign ore credible in the Great Lakes market.
- The threat was large enough to cast doubt on the survival of local producers.

8.2 Result 2: Productivity rose quickly after the shock

- Minnesota, U.S., and Canadian labor productivity increased sharply.
- Canadian materials productivity, capital productivity, and TFP also rose.
- The timing matches the crisis and subsequent work-rule changes.

8.3 Result 3: Within-mine improvement dominated selection

- Table 2 shows within-mine gains explaining most labor-productivity growth.
- Between-mine reallocation, exit, and entry do not explain the bulk of the productivity boom.
- This distinguishes the paper from a pure selection story.

8.4 Result 4: Work-practice changes explain the multi-input productivity pattern

- Looser work rules reduced overstaffing.
- They also increased machine operating time.
- This raises labor productivity, capital productivity, materials productivity, and TFP simultaneously.

8.5 Result 5: Standard input-substitution explanations are insufficient

- Relative factor prices and input ratios do not match the productivity timing closely enough.
- Capital deepening alone cannot explain materials productivity or TFP increases.
- The work-practice mechanism fits the full pattern better.

9 Short summary

- Schmitz studies the dramatic productivity recovery of U.S. and Canadian iron ore after the early-1980s crisis.
- Great Lakes producers had been protected by transport-cost advantages for nearly a century.
- Brazilian competition suddenly threatened the region when world steel-market developments changed delivered ore prices.
- Labor productivity roughly doubled within a few years.
- Canadian materials productivity rose by more than half, and Canadian TFP rose substantially.
- The main productivity gains came from changes in work practices, not simply new capital or exit of bad mines.
- Mine-level decomposition shows that within-mine productivity improvement explains most of the Minnesota productivity growth.
- Work-rule changes reduced overstaffing and raised the fraction of time equipment was in operating mode.
- This explains why labor, materials, and capital productivity could rise together.
- Figures 8 and 9 weaken standard relative-price substitution explanations.
- Figure 10 provides direct evidence from Minntac/USX that repair-hour intensity fell.
- The paper's broad message is that competition can raise productivity by forcing organizations to change internal practices.

10 Conclusion

10.1 Core conclusion

The paper concludes that competition and work practices are major determinants of productivity. The early-1980s foreign competition shock forced U.S. and Canadian iron ore producers to reorganize production. The resulting productivity gains were large, rapid, and visible across multiple input measures.

10.2 Economic intuition

Protected firms can operate with inefficient internal practices because market pressure is too weak to force costly bargaining and organizational change. Once foreign competition threatens survival, the cost of maintaining restrictive practices becomes too high. The firm, workers, and unions then have stronger incentives to renegotiate the organization of production. The same machines and workers can then produce more because labor is less overstaffed and capital spends more time operating.

10.3 What the paper teaches about productivity

The paper teaches that productivity is not only technology. It is also organization. Rules about who can perform tasks, how repairs are carried out, how many workers are required, and how downtime is handled can substantially affect measured productivity. These rules can remain inefficient for long periods in protected markets.

10.4 What the paper teaches about competition

Competition can improve productivity in at least two ways. First, it can select efficient producers and force inefficient producers to exit. Second, and more importantly in this paper, it can make existing producers change internal practices. The iron ore evidence shows that the second mechanism can be very large.

10.5 Policy implication

Trade protection, transport barriers, and other forms of insulation can preserve inefficient work practices. Reducing protection can generate productivity gains not only through cheaper imports but also through domestic reorganization. The transition may be disruptive, but the long-run productivity gains can be substantial.

10.6 Limitations

The paper is a detailed historical case study. It does not estimate a randomized treatment effect, and it cannot perfectly quantify every simultaneous managerial, technological, and bargaining change. Some data, especially for materials and capital in the U.S. industry, are imperfect. Still, the consistency of the timing, the decomposition, and the work-practice evidence makes the interpretation compelling.

10.7 Takeaway

The central takeaway is that competition can unlock productivity hidden inside firms. In the Great Lakes iron ore industries, the productivity revolution was not primarily about new technology or input substitution; it was about changing work practices that had kept existing resources from being used efficiently.